Vardan Gupta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.34
Kiddy's Corner Hr. Sec. School, Gwalior	Class XII (CBSE)	2023	93.60%
Kiddy's Corner Public School, Gwalior	Class X (CBSE)	2021	96.60%

WORK EXPERIENCE

Flipkart | Software Development Engineer (SDE) Intern (Bangalore, India | May 2026 - July 2026)

Project-1: API Test Automation | Java, ZeroCode TDD, Gradle

- Constructed an API test automation framework using ZeroCode to replace manual QA, **reducing regression testing time** by **96%** (from 3 hours to <7 minutes) with a **0%** false failure rate.
- Integrated the automation suite into the CI/CD pipeline, establishing a continuous testing framework that guaranteed deployment stability across **80%** of **backend environments.**
- Automated the generation and delivery of detailed HTML telemetry reports and trace logs directly to the team's Gmail, ensuring a **100% success rate** while drastically accelerating issue detection and streamlining system debugging.

Project-2: LLM-Powered Alert Analysis | Spring AI, Ollama, Spring Boot, Vite, Cassandra, Redis

- Architected an autonomous AI alert triage system integrating local LLMs to classify production incidents, saving on-call teams **15+ hours** of manual log review weekly.
- Designed a custom asynchronous HTTP transport protocol for live telemetry ingestion, ensuring 100% data privacy and **zero API costs.**
- Implemented a live analytics dashboard backed by a Redis/Cassandra caching layer to evaluate **over 500 daily incident data points** in real time and minimize manual triage latency.

PROJECTS

IIT-H Complaints Website (Group Project) | Next.js, Firebase, Firestore, Tailwind CSS | Github

- Programmed a scalable full-stack facility management platform serving **500+ active users**, streamlining campus issue resolution and tracking.
- Integrated Google OAuth 2.0 for secure authentication to manage **200+ monthly logins**, and leveraged Firestore to deliver real-time state synchronization.
- Automated complaint routing and email notifications using **Google Apps Script**, achieving a 100% delivery rate and significantly reducing response latency.

Battery Degradation Modeling (Academic Research Project) | Grade: A | Python, PyTorch | Research Report

- Designed a Multi-Task LSTM incorporating physical degradation embeddings and thermodynamic penalties, achieving a **0.9995 R-squared** score for State of Charge (SOC) prediction and reducing Mean Squared Error (MSE) to **4.27e-5.**
- Devised a cascaded CNN decoder to generate full 520-step voltage sequences in a single forward pass, eliminating standard RNN autoregressive bottlenecks (which stalled at a **0.863 R-squared** score) and improving baseline accuracy.
- Constructed a State of Health (SOH) degradation tracking framework utilizing Ridge Regression, capturing near-linear aging trajectories across 1,000 discharge cycles to achieve a **0.9976 R-squared** score and an MSE of **3.55e-9.**

Abstractive Text Summarization Pipeline | Python, PyTorch, FastAPI, Hugging Face | Github

- Created an end-to-end ML pipeline and FastAPI inference service for conversational text, delivering real-time abstractive summaries with a **1.2s average inference latency.**
- Fine-tuned a Pegasus sequence-to-sequence transformer on the **16,000-example SAMSum dataset**, implementing custom attention masking and beam-search generation to handle complex inputs.
- Optimized training workflows utilizing gradient accumulation for GPU memory efficiency, achieving highly accurate

29.2 ROUGE-1 and **22.6 ROUGE-L** scores against human baselines.

SKILLS

- **Languages:** C/C++, Java, Python, JavaScript, HTML, MATLAB, Google Apps Script
- **Web & Backend:** React.js, Next.js, Node.js, Express.js, Spring Boot, FastAPI, Tailwind CSS, Vite
- **AI & Data Science:** TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, Transformers, NLTK, Spring AI, Ollama
- **Databases & Tools:** Git, Docker, Cassandra, Redis, Firebase, Firestore, Gradle, OAuth 2.0, VS Code, ZeroCode
- **Core Concepts:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Machine Learning, Deep Learning

RELEVANT COURSES

- **Machine Learning & AI:** Artificial Intelligence, Computer Vision in Chemical and Life Science
- **Mathematics & Statistics:** Linear Algebra, Probability Theory, Applied Statistics, Principles of Game Theory, Numerical Methods, Applied Mathematics in Chemical Engineering
- **Programming & Computation:** Introduction to Programming, Business Analytics using Python

SCHOLASTIC ACHIEVEMENTS

Director's Award of Academic Excellence: For securing the highest CGPA in the department in AY 2023-24.

Demonstrated advanced programming and analytical capabilities (using C++) through competitive programming:

- **Codeforces:** Specialist (Max Rating: 1424)
- **LeetCode:** Knight (Max Rating: 1908)
- **CodeChef:** 3-Star (Max Rating: 1691)
- **AtCoder:** 5 Kyu (Max Rating: 1093)